

FULL TEST - 1

CBSE BOARD PATTERN

1.

(4) b and d

2.

(4) All of these

3.

(1) Tallness is dominant over dwarfness

4.

(3) b, c

5.

(3) translocation

6.

(1) Arteries

7.

(2) a Convex mirror

8.

(3)

9.

(1) Scattering of light is not enough at such heights

10.

(2) Refraction, dispersion and total internal reflection

11.

(2) high resistance and low melting point

12.

(2)

13.

(3) Pattern of magnetic field associated with the solenoid is different from the pattern of the magnetic field around a bar magnet.

14.

(1) an exothermic reaction

15.

(1) CO_2

16.

(2) 2

17.

(4) (i) and (iii)

18.

(4)

19.

(3)

20.

(2) 148

21. The raw materials required for photosynthesis are sunlight, CO₂ and water.

Plant gets sunlight from the sun, CO₂ from the air through the stomata and water from roots after absorption from soil.

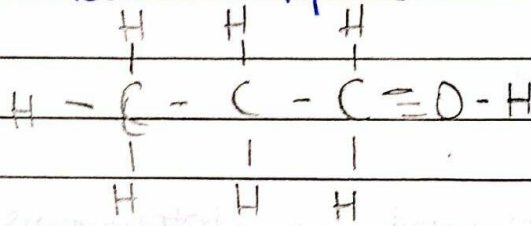
22. Haemoglobin is responsible for transporting O₂ to different parts of the body. A deficiency in haemoglobin would lead to lower levels of O₂ in our body. This would decrease our efficiency.

23. A current carrying conductor will experience no force if the direction of current is parallel to the direction of the magnetic field.

24. Three characteristics of the image formed by the plane mirror are:

- 1) It is the same size as of the object
- 2) It is virtual and erect
- 3) It is laterally inverted

25. The name is Propanal



26.

(i) Magnesium

(ii) Copper

27. To maximise the area for exchange of gases, lungs have following features:

- 1) They have a large surface area for greater absorption of gases
- 2) They have a thin membrane which allows easy exchange of gases
- 3) The presence of millions of balloon like structures called alveoli also increases area for exchange of gases
- 4) Lungs have a good network of blood vessels which help in transporting absorbed gases

28.

(a) Liver - Liver produces Bile juice and stores it in gall bladder. Bile helps in digestion of lipids.
Pancreas - Pancreas secrete ~~import~~ pancreatic enzymes which help in digesting food.

(b) Nutrition - It is the process of taking in food and converting it into energy and other nutrients vital for life.

Nutrients - It is a substance used by an organism to survive, grow. Ex: proteins, fats etc.

Holozoic nutrition

1) It involves ingestion, digestion, absorption to produce food energy.

2) Food is digested within the body.

3) Ex: Fungi, Bacteria

Saprophytic nutrition

1) Organisms obtain their food from dead and decaying plants and animals.

2) Food is digested outside the body.

3) Ex: Humans, Amoeba

29. Differences between self pollination and cross pollination are :

Self Pollination
 1) Transfer of pollen ~~grains~~ grains from anther to Stigma of same flower or different flower in the same plant

Cross Pollination
 1) Transfer of pollen grains from anther of one flower to the stigma of another flower in on another plant

2) Carried out by bees,

2) Carried out by water, wind,

3) Leads to less variety

3) Leads to greater ~~variety~~ variety

The various agents which help in pollination are wind, water, animals.

30. According to Joule's law of heating, when a current is passed through a conductor for a certain amount of time, heat produced is given by :

$$H = I^2 RT$$

where H is heat, I is current, R is resistance and T is time.

$$l = 12m$$

$$r = 2 \times 10^{-4} m$$

$$\rho = 3.14 \times 10^{-8} \Omega m$$

$$R = ?$$

$$R = \frac{\rho l}{A}$$

$$R = \frac{3.14 \times 10^{-8} \times 12}{(2 \times 10^{-4})^2 \times 3.14}$$

$$= \frac{3.14 \times 10^{-8} \times 12}{4 \times 10^{-8} \times 3.14}$$

$$= \frac{10^{-8} \times 3}{10^{-8}}$$

$$= 3$$

$$R = 3 \Omega$$

$$R = 3 \Omega$$

31

(i) Power for distant vision = -0.5 D

We know that,

$$P = \frac{1}{b(\text{m})}$$

$$-0.5 = \frac{1}{b}$$

$$b = \frac{-1 \times 10^2}{0.5}$$

$$b = -2 \text{ m or } -200 \text{ cm}$$

(ii) Power for near vision = +1.5 D

We know that,

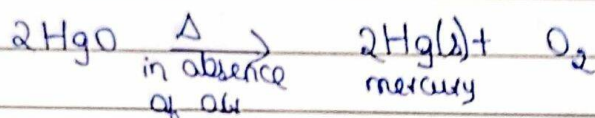
$$P = \frac{1}{b(\text{m})}$$

$$b = \frac{1}{P}$$

$$b = \frac{1 \times 10^2}{1.5} = 0.66 \text{ m}$$

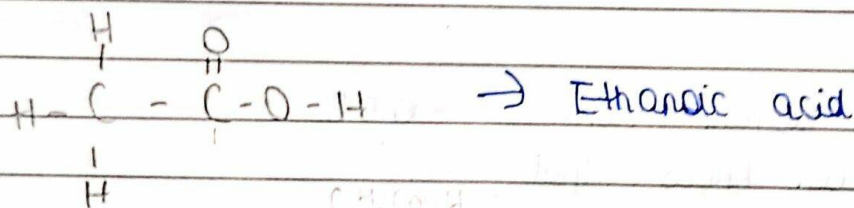
$$b = 0.66m$$

32. The metal is mercury and its Ore is Cinnabar (~~HgS~~)

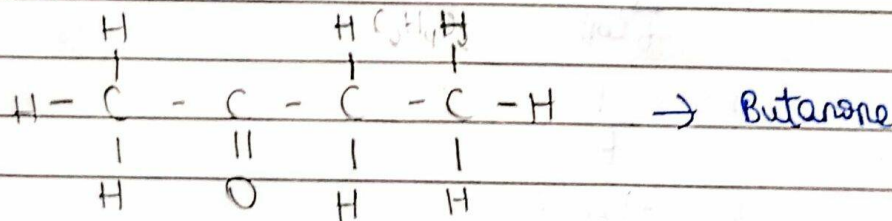


33.

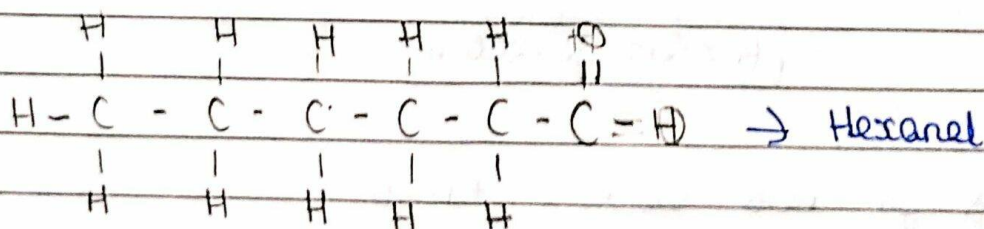
(i)



(ii)

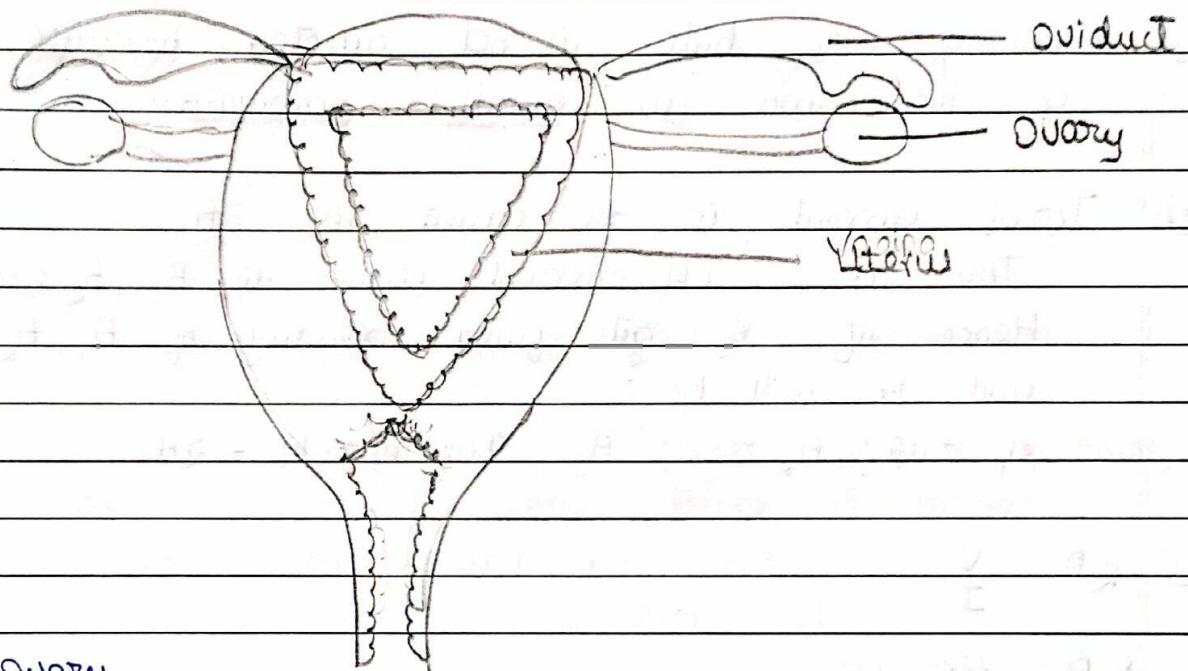


(iii)



34

(i)



(a) Ovary

(b) Oviduct

(c) Uterus

(ii) The following events occur when egg is not fertilised :

- 1) The thick and spongy lining of the uterus breaks down along with the blood vessels
- 2) It comes out through the vagina as menstrual blood
- 3) This process is known as menstruation

35.

(i) Glow of other bulbs is not affected because all of them are in parallel combination.

(ii) Total current in the circuit is 3A.

This means 1A current flows in B_1 , B_2 and B_3 .
Hence if B_2 gets fused reading of A_1 , A_2 , A_3 and A will be:

$A_1 = 1A$; $A_2 = 0$; $A_3 = 1A$ and $A = 2A$

(iii) $R = \frac{V}{I}$

$$R = \frac{4.5}{3} = 1.5$$

$$R = 1.5 \Omega$$

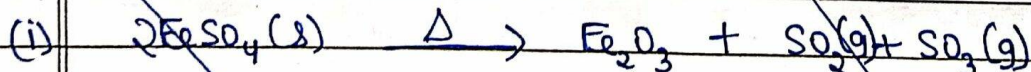
$$P = \frac{V^2}{R}$$

$$= \frac{4.5 \times 4.5}{1.5}$$

$$P = 13.5 W$$

36.

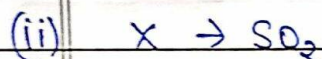
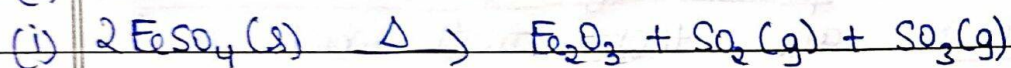
(a)



(ii)

36.

(a)



(iii) This reaction is a decomposition reaction

(b) Zinc is a reactive metal and lies above Hydrogen in the reactivity series. Hence it displaces hydrogen from hydrochloric acid.



Copper lies below hydrogen in the reactivity series. Therefore it cannot displace hydrogen from hydrochloric acid.



37.

(i) Mendel selected pea plant because of the following reasons

- 1) It has a short life cycle
- 2) It can be cross pollinated or self pollinated
- 3) It has some pairs of differentiating characters

(ii) Dominant traits are those traits which are expressed even in the presence of an alternative trait.
Recessive traits are those traits which are expressed only in the presence of a similar trait.

(iii) Two visible traits of garden pea that Mendel considered in his experiments are :

- 1) Pea plants were either tall or dwarf
- 2) Colour of the seeds was yellow or green

38.

(a) The range of wavelength of visible light spectrum is from 380nm to 780nm

(b) Different wavelengths of energy produced by light source when it passes through prism because different colours of light have different speeds. Speed is directly proportional to velocity.

(c) Different colours of light have different speeds. Since speed is directly proportional to ^{wavelength} ~~velocity~~, more the velocity, more the wavelength and lesser the dispersion of light.
This is the cause of dispersion of light.

39



(ii) Anode $\rightarrow$ chlorine, Cathode $\rightarrow$ Hydrogen

(iii) Chlorine is used in the manufacturing of bleaching powder.
Hydrogen is used in rocket fuels.